PRIMATE GENOMICS

WEEK 45 PAPER DISCUSSION

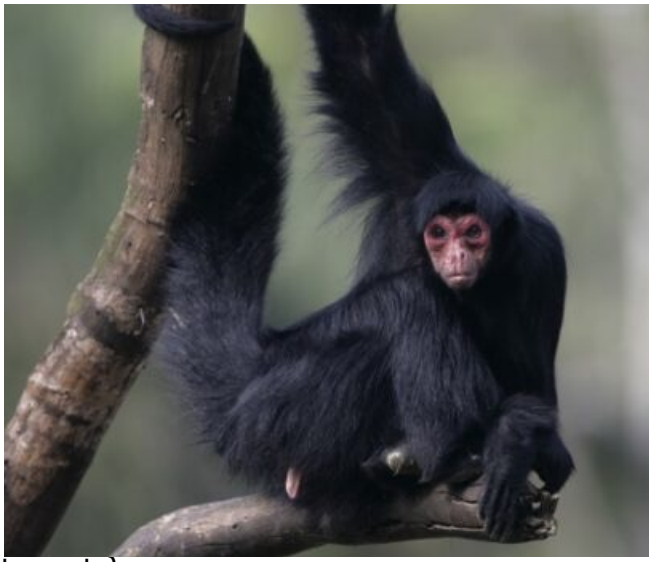
DEPARTMENT OF MOLECULAR BIOLOGY AND GENETICS

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EVOLUTIONARY THINKING
29. AUGUST 2025

EMMA FEMKE DIEPEVEEN
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PRIMATE OF THE WEEK



Guiana Spider monkey

Ateles paniscus

Aka black or red-faced spider monkey

Northern South America in the high rainforest
Spend their entire live basically 25m from the ground

Their long tail serves as an extra limb (can just hang on it freely while using their hands)
It is 2/3 of its length and has a hairless tip that has skin with fine ridges (like a human fingerprint) forming a friction pad.

Their fur is long, silky and glossy black that glimmer in the sunlight (twilight vibe)



Use their hands as hooks (thumb is reduced or nearly absent)

When annoyed, they might shake branches, jump around or toss twigs.

Hold hands when they meet after time apart

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TODAY

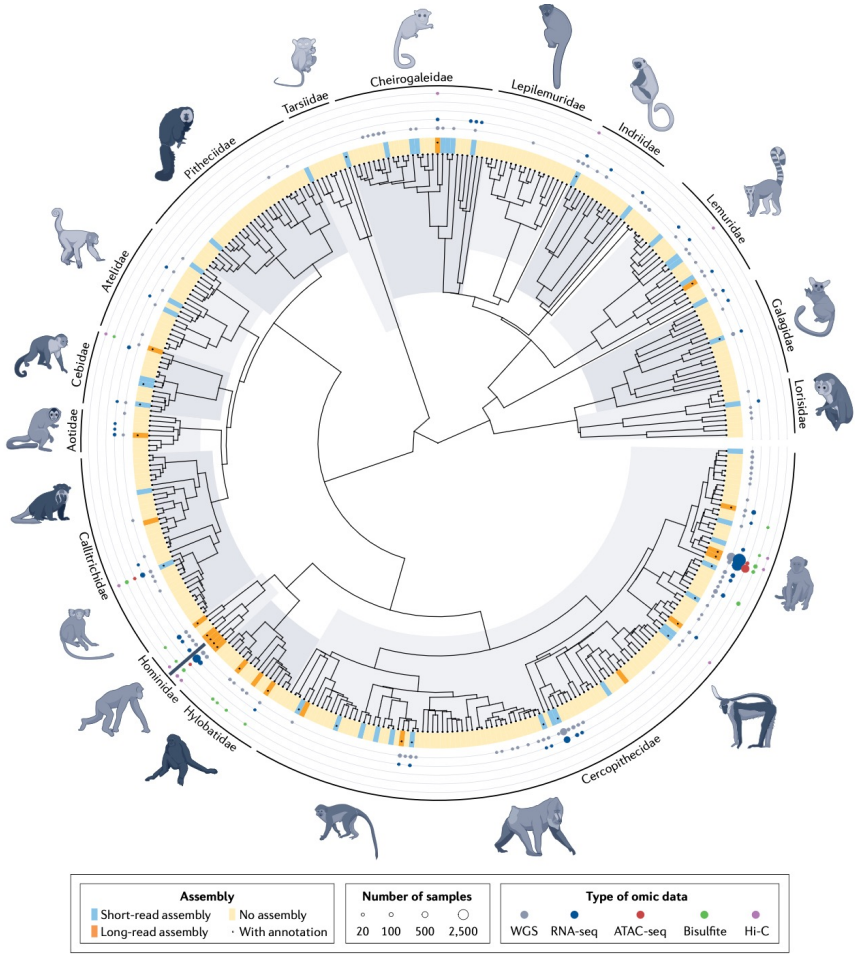
Paper discussions of

- Kuderna et al. (2023) A global catalog of whole-genome diversity from 233 primate species. Science 380: 906–913.

- Juan et al. (2023) Current advances in primate genomics: novel approaches for understanding evolution and disease. Nature Reviews Genetics.

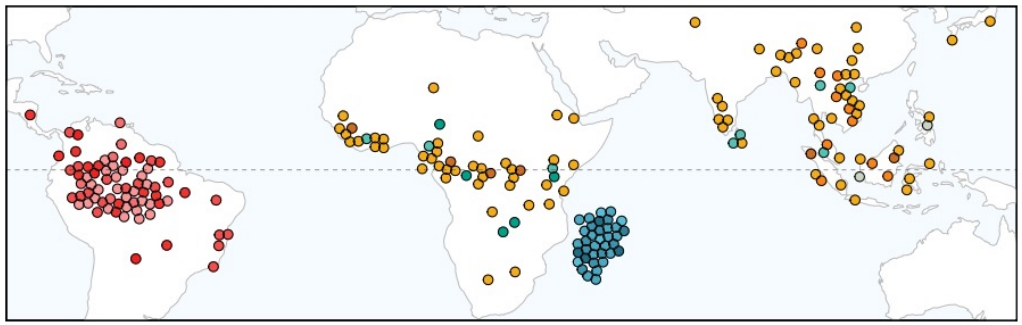
WHY STUDY PRIMATES?

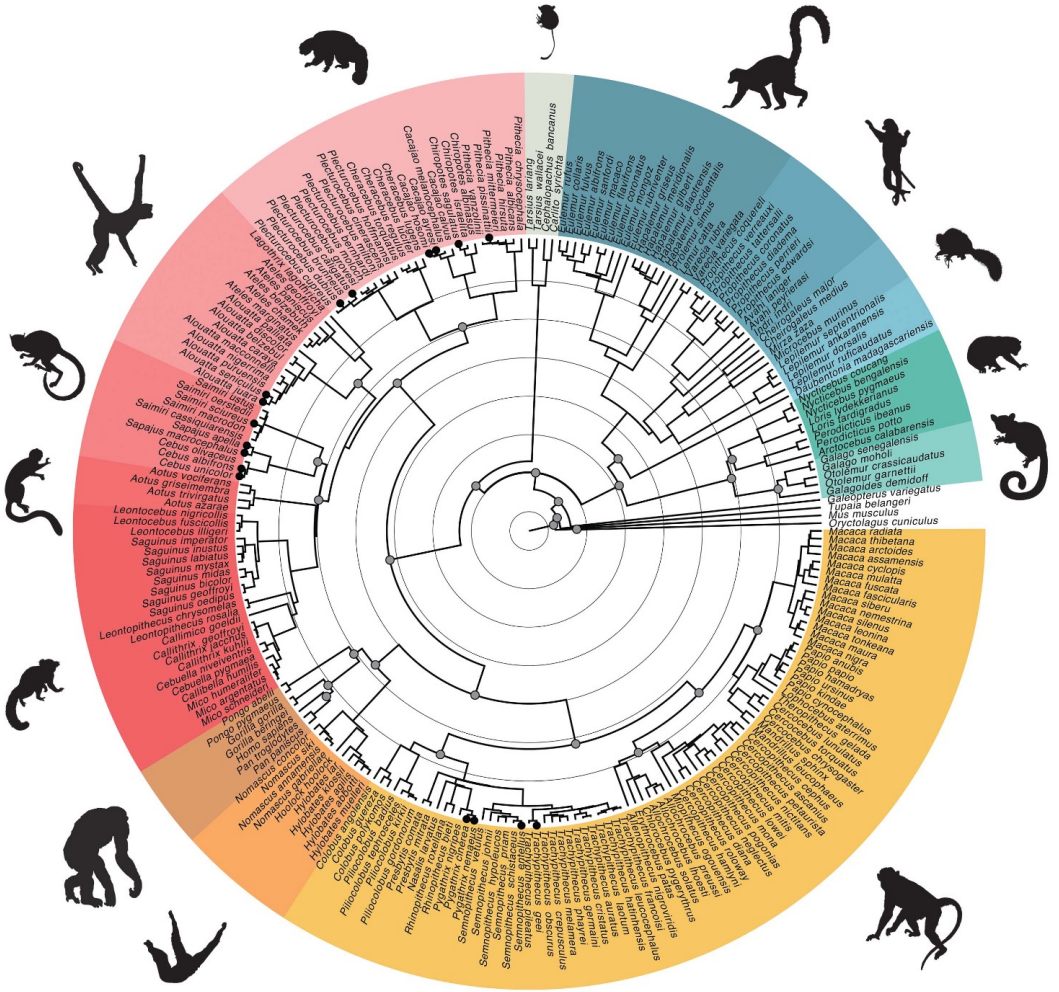
- More than 500 species, but genomic data exist for only a handful
- Genomes are very similar to each other, and species barriers are often incomplete.
- Many species are threatened (60 %).



MEET THE DATA: THE PRIMATE TREE (KUDERNA 2023)

809 genomes from 233 species, all 16 families, mostly wild-born animals.





NUMBERS YOU CAN GET FROM A GENOME

Heterozygosity (π): long-term *N*e.

Runs of homozygosity: recent inbreeding.

Missense/synonymous ratio: how efficient selection is.

π looks thousands of generations back, while long RoH look only a few generations back.

MUTATION RATE: PER GENERATION VERSUS PER YEAR

short-lived species



(e.g. a lemur, generation time about 4 years)

Few germ-cell divisions before reproducing

→ fewer mutations **per generation**

But many generations per year

→ more mutations **per year**

Long-lived species



(e.g. a great ape, generation time about 20 years)

Many germ-cell mutation before reproducing

→ more mutations **per generation**

But few generations per year

→ less mutations **per year**

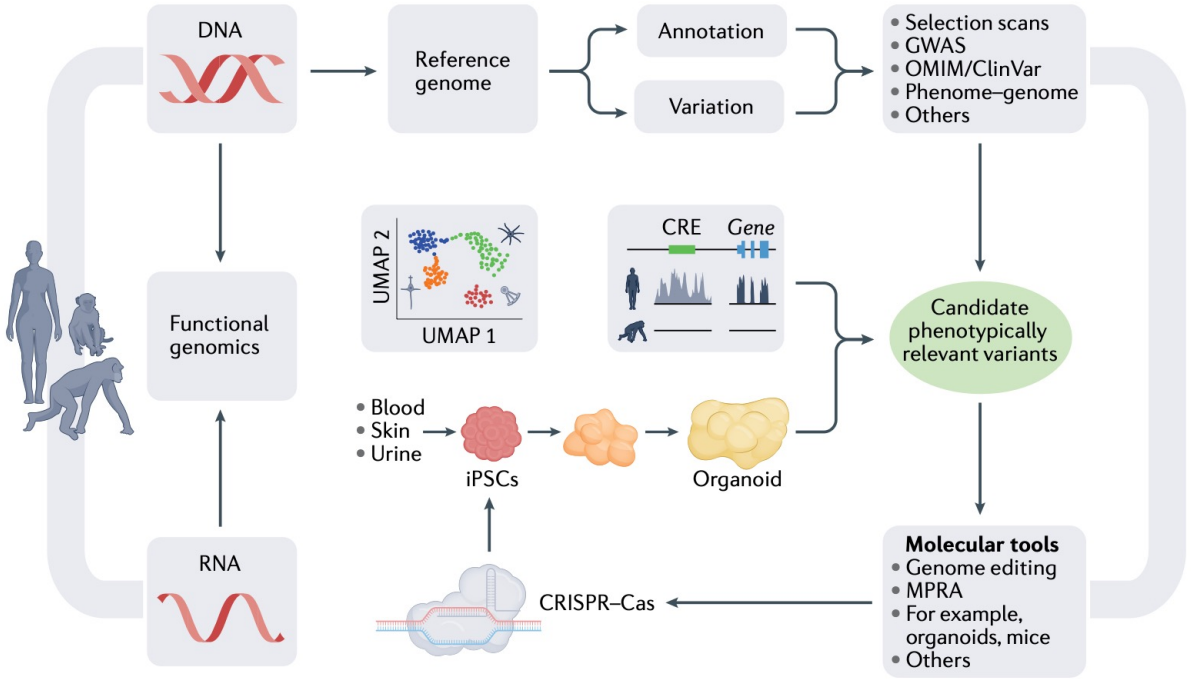
FROM GENOME TO FUNCTION (JUAN ET AL)

Comparing genomes gives candidates, but which differences actually matter?

Genomes: reference genomes and diversity for many species

Candidates: variants that differ between species or are under selection

Testing: do these variants change how the gene works



TODAY

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